

Development of a customizable Enterprise Resource Planning on frappe framework for Human Resource management with event-driven blockchain integration

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ABSTRACT

The central challenge in modern enterprise management is the "Super-user Paradox," where privileged administrators can silently manipulate data and delete audit logs within centralized, mutable databases like MariaDB. To address this vulnerability in Human Resource Management (HRM), we developed a customizable architecture on the Frappe Framework that anchors ERPNext records to an immutable Hyperledger Besu blockchain via an event-driven Change Data Capture (CDC) pipeline using Debezium and Apache Kafka. Experimental results show that the DB-to-Kafka stage achieves sub-second latency ($\approx 0.82s$) and the streaming infrastructure introduces negligible overhead ($\approx 1ms$), though the primary bottleneck remains the consensus-heavy Consumer-to-Blockchain stage ($\approx 2.0s$ per record). While individual records synchronize in approximately 2.83 seconds, bulk operations scale linearly, with 250 records requiring 128 seconds. Performance analysis further indicates that an optimal 4 - 6 validator node configuration achieves 42 - 50 TPS with 99.2 - 99.7% success rates, effectively bridging the gap between flexible ERP operations.

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1. INTRODUCTION

Enterprise Resource Planning (ERP) systems have become the digital backbone of modern organizations, integrating functions such as finance, supply-chain, and human-resources into a unified platform [1], [2]. These systems consolidate data from multiple departments, enabling real-time visibility into organizational operations and supporting data-driven decision-making across all levels of management. While open-source solutions like ERPNext have dramatically lowered the entry barrier for adopting ERP capabilities by eliminating expensive licensing fees and offering extensive customization possibilities [3], [4], their reliance on a centralized, mutable relational database managed by privileged administrators introduces a critical vulnerability known as the Super-user Paradox [5]. In this scenario, the very entity responsible for safeguarding the system possesses the technical ability to alter transactional data and tamper with audit logs without detection, thereby jeopardizing data integrity and regulatory compliance. This architectural weakness undermines the foundational trust that organizations place in their ERP systems, particularly when sensitive employee and financial data are involved.

In the domain of Human-Resource Management (HRM), the trustworthiness of attendance records is paramount because these records directly influence payroll calculations, performance evaluations, and compliance with labor regulations. Recent advances in IoT-enabled smart attendance technologies ranging from RFID-based readers [6], [7], [8] to facial-recognition systems [9], [10] and speech-recognition interfaces [11] have greatly improved data capture efficiency by automating the clock-in and clock-out process and reducing human error. Researchers have also explored advanced solutions such as smart classroom attendance management [12] and Industry 4.0-oriented monitoring technologies [13]. However, these solutions typically persist their data in conventional SQL databases, leaving them exposed to unauthorized "manager overrides" where a privileged user can retroactively modify attendance entries [14]. Such manipulations can facilitate payroll fraud through inflated overtime claims, undermine employee trust when discrepancies are discovered, and violate labor regulations that mandate accurate record-keeping.

To address these challenges, we propose an architecture that couples the Frappe/ERPNext framework with a private Hyperledger Besu blockchain via an event-driven Change-Data-Capture (CDC) pipeline. By leveraging Debezium to capture row-level changes from the MariaDB database and streaming these change events into Apache Kafka, the system decouples the high-frequency transactional layer from an immutable evidentiary layer stored on the blockchain [15]. Hyperledger Besu was selected for its enterprise-grade performance, support for private transactions, and compatibility with Ethereum tooling [16]. This design ensures that every attendance event whether captured by RFID readers, facial-recognition cameras, or speech-recognition terminals is recorded immutably on the blockchain, providing cryptographic proof of authenticity and tamper-evidence that can withstand forensic scrutiny.

The integration offers several key benefits that distinguish it from existing approaches. In terms of data trust and non-repudiation, the blockchain's append-only ledger guarantees that once an attendance record is written, it cannot be altered or deleted without detection, thereby eliminating the Super-user Paradox [5]. Regarding auditability and compliance, the immutable logs simplify regulatory reporting and internal audits by providing a verifiable chain of custody for every data point, satisfying legal requirements for traceability in payroll and labor law enforcement [17]. With respect to scalability and real-time processing, the CDC-Kafka pipeline enables near-real-time propagation of attendance events to the blockchain without degrading ERPNext's transactional performance, as Kafka's distributed architecture can handle high-throughput message streams [15]. Concerning extensibility, the modular event-driven approach allows future integration of additional IoT sensors such as smart building telemetry [18], [19], mixed-reality interfaces [20], [21], [22], and advanced analytics engines while preserving data integrity across all connected systems.

Table 1 presents a comparison of the proposed system with previous works across key ERP-related features. The comparison illustrates that while prior works have explored ERP implementation methodologies using platforms like Odoo [1], blockchain integration for supply chain operations on ERPNext [23], blockchain-enabled hospital information system management [17], and conceptual frameworks for blockchain-ERP coupling on proprietary systems [5], none of them combine an open-source ERP platform with IoT-enabled attendance capture and a real-time CDC pipeline specifically targeting HRM attendance management. Our proposed system uniquely integrates all of these capabilities within the Frappe/ERPNext ecosystem, delivering end-to-end tamper-proof attendance records.

Table 1. Comparison with Previous Works

Feature	Gonzalez et al. [1]	Zainuddin et al. [23]	Satrio et al. [17]	Tricase et al. [5]	Proposed
Platform Type	Odoo	ERPNext	Custom HIS	SAP/Oracle	ERPNext
Blockchain Integration	No	Yes	Yes	Yes	Yes
IoT Integration	No	Yes	No	No	Yes
Real-Time Data Streaming (CDC)	No	No	No	No	Yes
Data Immutability	No	Partial	Yes	Yes	Yes

Target Domain	General ERP Implementation	Supply Chain	Hospital Information System	Generic ERP-Blockchain	HRM & Attendance Management
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By embedding blockchain as a trusted evidentiary layer beneath ERPNext's transactional engine, the proposed system delivers a robust solution to the longstanding problem of data trust in HRM attendance management. This architectural intervention not only protects against internal fraud but also positions organizations to meet increasingly stringent data-governance requirements, paving the way for secure, transparent, and auditable enterprise operations.

2. METHOD

The proposed system follows a microservices-oriented architecture comprising three main layers: the ERP Platform (Application Layer), the Event Streaming Layer (Middleware), and the Blockchain Network (Security Layer).

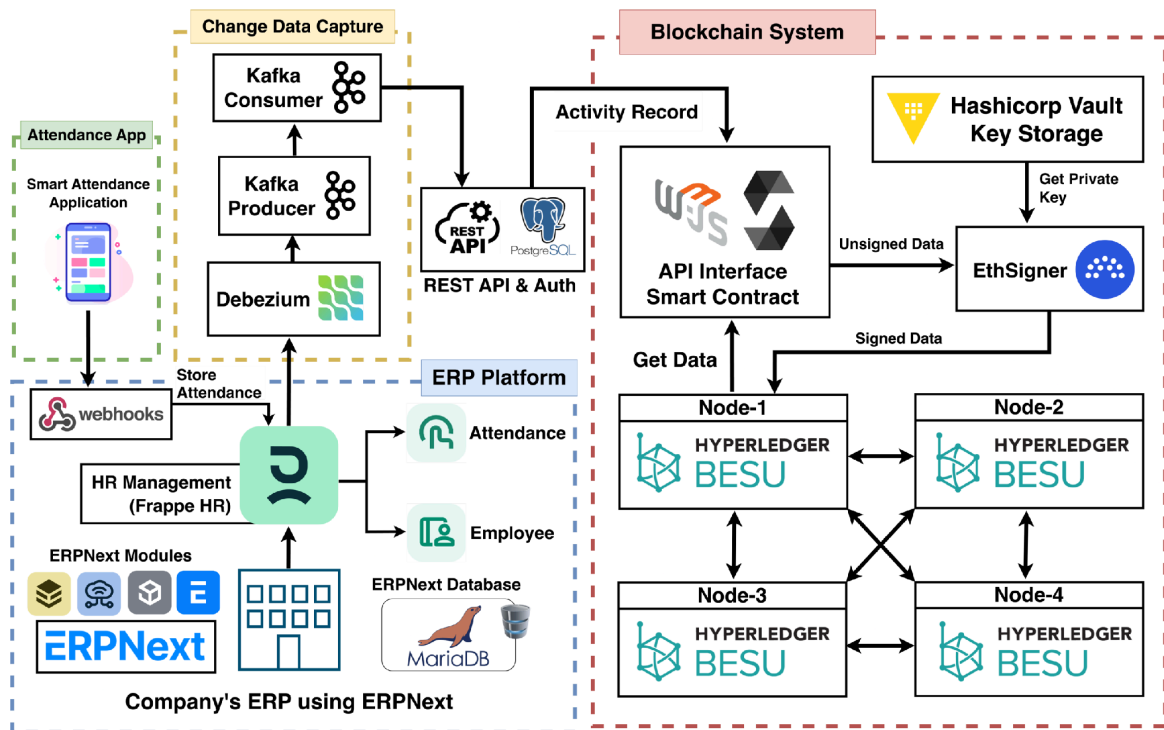


Figure 1. Proposed Design System

2.1. ERPNext

ERPNext is an open-source, modular Enterprise Resource Planning solution built on the Frappe Framework, selected for this study due to its comprehensive domain coverage, zero licensing costs, and highly customizable architecture that supports rapid extension development. Unlike proprietary ERP systems such as SAP or Oracle, ERPNext provides full access to its source code and a well-documented API layer, enabling researchers and developers to implement custom integrations without vendor lock-in. For this study, ERPNext is deployed using Docker containers orchestrated via Docker Compose, which ensures consistent development and deployment environments across different stages of the system lifecycle. The containerized architecture includes dedicated services for the backend application server, frontend Nginx reverse proxy, Redis instances for caching and background job queuing, and a scheduler for periodic task execution. The system utilizes MariaDB version 10.6 as the primary transactional database, where all day-to-day HR operations including employee onboarding, attendance logging, and leave management are stored initially. To enable Change Data Capture (CDC), the MariaDB container is configured with specific binary logging parameters including row-based binlog format and full row image capture, which provide the granular change events required by Debezium for downstream processing [24].

2.2. Frappe Framework

The Frappe Framework is the full-stack web application framework upon which ERPNext is built, providing a metadata-driven development paradigm that enables rapid customization and extension of ERP functionality [25]. Frappe employs a DocType system where data models are defined as JSON-based schemas that automatically generate database tables, REST APIs, and user interface forms without requiring manual coding. Since standard ERP modules lack native support for IoT device management and blockchain integration, we leveraged Frappe's extensibility to develop two custom applications: IoT and Blockchain. The IoT app introduces DocTypes for managing IoT devices and sensor data, and includes a custom Workspace that provides a pre-configured dashboard for embedding Home Assistant an open-source home automation platform used in our previous work [26] directly within the ERPNext interface, enabling HR administrators to monitor smart attendance devices and environmental sensors from a unified dashboard. The Blockchain app provides DocTypes for registering smart contracts and tracking blockchain transactions, and includes a pre-configured Workspace for embedding the blockchain explorer used to monitor transaction confirmations and block finality on the Hyperledger Besu network. Both apps are installed using the standard Frappe bench commands and integrate seamlessly with ERPNext's existing modules, demonstrating that the platform can be extended to support emerging technologies without modifying its core codebase a critical advantage for maintaining upgrade compatibility with future releases [27].

2.3. Human Resource Management

By default, ERPNext does not include a dedicated Human Resource Management module; therefore, we installed Frappe HR (formerly ERPNext HRMS), an official open-source application that extends ERPNext with comprehensive HR functionality including employee lifecycle management, leave and attendance tracking, payroll processing, and recruitment workflows. The HR module is the focal point of this study, as attendance and employee data represent high-value targets for manipulation in payroll fraud scenarios. The customization in this research focuses on two key entities that form the foundation of HRM operations.

2.3.1. Employee

The Employee DocType serves as the master data entity within Frappe HR, storing critical fields such as *employee_name*, *date_of_joining*, *department*, *designation*, *company*, and *default_shift*. We extended the standard model to include immutable referencing by leveraging the CDC pipeline to capture any creation or modification of an employee profile. When changes occur, the creation and modified timestamps in the MariaDB binlog are detected by Debezium, enabling the blockchain consumer to compute a cryptographic hash of the record and anchor it on the Hyperledger Besu network. This mechanism ensures that the identity of the workforce cannot be quietly altered without generating an immutable audit trail.

2.3.2. Attendance

Attendance data is high-volume and time-sensitive. Frappe HR provides two related DocTypes: Employee Checkin for capturing raw check-in/check-out events with fields such as *employee*, *log_type* (IN/OUT), *time*, and *device_id*, and Attendance for the finalized daily attendance record with fields including *attendance_date*, *status* (Present/Absent/On Leave/Half Day), *in_time*, *out_time*, *working_hours*, *late_entry*, and *early_exit*. In our implementation, attendance data from external smart attendance systems using face recognition is received via Frappe's webhook endpoints, which create Employee Checkin records [28]. Frappe HR then automatically processes these checkins to generate the corresponding Attendance records. The CDC pipeline captures these database changes, and the consumer service sends the data to the blockchain via API. Unlike standard implementations that treat attendance records as mutable database entries, our system ensures that once an attendance log is committed to the ERP, its cryptographic hash is anchored on the blockchain, preventing retroactive changes to working hours.

2.4. Blockchain System

We employed Hyperledger Besu, an enterprise-grade Ethereum client developed under the Linux Foundation's Hyperledger umbrella, to establish a private permissioned blockchain network [29], [30]. Besu was selected for its support of diverse consensus mechanisms (including IBFT 2.0, QBFT, and Clique), privacy features suitable for corporate environments, and full compatibility with Ethereum tooling and smart contract development workflows.

2.4.1. Hyperledger Besu

The network is configured as a private consortium chain using the IBFT 2.0 (Istanbul Byzantine Fault Tolerant) consensus mechanism, which provides immediate transaction finality and resistance to Byzantine failures. Unlike public blockchain networks where transaction fees and confirmation times are unpredictable, the private Besu network offers deterministic performance characteristics essential for enterprise integration. The consortium model allows the organization to control which nodes can participate in the network through a permissioning system, ensuring that sensitive HR data hashes are not exposed on a public ledger. Each participating node maintains a local copy of the blockchain, enabling distributed verification while preserving data sovereignty within organizational boundaries.

2.4.2. Node Validator

The network consists of multiple Validator Nodes responsible for verifying transactions and assembling blocks according to the IBFT 2.0 consensus protocol. In this configuration, validators take turns proposing blocks in a round-robin fashion, with each block requiring approval from a supermajority ($2f+1$ out of $3f+1$ validators, where f is the maximum number of faulty nodes tolerated). The performance implications of the number of validators are analyzed in Section 3.3, where we examine the trade-off between decentralization (more validators) and transaction throughput (fewer consensus rounds). For the prototype implementation, we deployed a four-node validator network, providing fault tolerance for one Byzantine node while maintaining sub-second block times.

2.4.3. Smart Contract

A Solidity-based Smart Contract was developed to store the cryptographic hashes of ERP records on the blockchain. The contract implements a mapping structure that associates ERPNext's internal record identifiers (Primary Keys such as HR-EMP-00001 for employees and HR-ATT-2024-00001 for attendance) with their corresponding data hashes, timestamps, and metadata. Key functions include *storeEmployee()* for anchoring employee master data, *storeAttendance()* for anchoring attendance events, and *verifyRecord()* for validating that a given data payload matches the stored hash. The contract acts as the immutable "source of truth," enabling any authorized party to independently verify that an ERP record has not been tampered with since its original creation. Events are emitted upon each storage operation, providing an indexed log that facilitates efficient querying of historical anchoring transactions.

Algorithm 1. Smart Contract: Document Employee & Attendance

```

Data: Initialize
recordId: string indexed,
createdTimestamp: uint,
modifiedTimestamp: uint,
modifiedBy: string,
allData: string,

Mapping: records[recordId] → Record
Array: recordIds[]
Mapping: recordExists[recordId] → boolean
Result: Stored recordId, createdTimestamp, modifiedTimestamp, modifiedBy, allData
Event dataEmployee (recordId: string, createdTimestamp: uint, modifiedTimestamp:
    uint)
Event dataAttendance (recordId: string, createdTimestamp: uint, modifiedTimestamp:
    uint)

Function storeRecord(recordId, createdTimestamp, modifiedTimestamp, modifiedBy,
    allData):
    isNew ← NOT recordExists[recordId];
    Set records[recordId] ← {recordId, createdTimestamp, modifiedTimestamp,
        modifiedBy, allData};
    IF isNew THEN
        recordIds.push(recordId);
        recordExists[recordId] ← true;
        Emit dataEmployee OR dataAttendance event with (recordId, createdTimestamp,
            modifiedTimestamp);

```

```

END IF
Return true;

```

The smart contract storage mechanism can be formally represented as follows:

$$D = \{r, t_c, t_m, u, \delta\}$$

Where:

- D = represent a document (Employee or Attendance)
- r = recordId (unique identifier, e.g., HR-EMP-00101, HR-ATT-2025-01223)
- t_c = createdTimestamp (Unix timestamp of creation)
- t_m = modifiedTimestamp (Unix timestamp of last modification)
- u = modifiedBy (user identifier who made the change)
- δ = allData (serialized JSON containing all document fields)

The storage operation is defined as:

$$S: D \rightarrow B$$

$$S(D) = \{INSERT(D) \rightarrow emit E_{stored} \text{ if } \nexists M[r] \text{ } UPDATE(D) \rightarrow emit E_{updated} \text{ if } \exists M[r]\}$$

Where:

- M = mapping structure storing records by recordId
- E_{stored} = event emitted for new records
- $E_{updated}$ = event emitted for updated records

The blockchain guarantees immutability such that for any stored document D at block height h :

$$\forall h' > h: H(D_{h'}) = H(D_h)$$

Where H represents the cryptographic hash function, ensuring that once recorded, historical states cannot be altered without detection.

Table 2. Document Employee & Attendance Structure Comparison

Attribute	Employee Document	Attendance Document
Record ID	HR-EMP-00XXX	HR-ATT-YYYY-XXXXX
allData	{employee_name, gender, date_of_birth, department, designation, ...}	{employee, attendance_date, status, in_time, out_time, shift, ...}

Both contracts implement identical storage logic, enabling consistent data integrity verification across all HRM document types while maintaining flexibility through the JSON-serialized `allData` field.

2.5. Change Data Capture

To avoid modifying the core ERPNext codebase which would complicate future version upgrades and violate best practices for maintainable software architecture we implemented an external Change Data Capture (CDC) pipeline that operates at the database level [31]. This approach ensures that all data mutations are captured regardless of whether they originate from the ERPNext user interface, API calls, direct SQL access, or batch import operations.

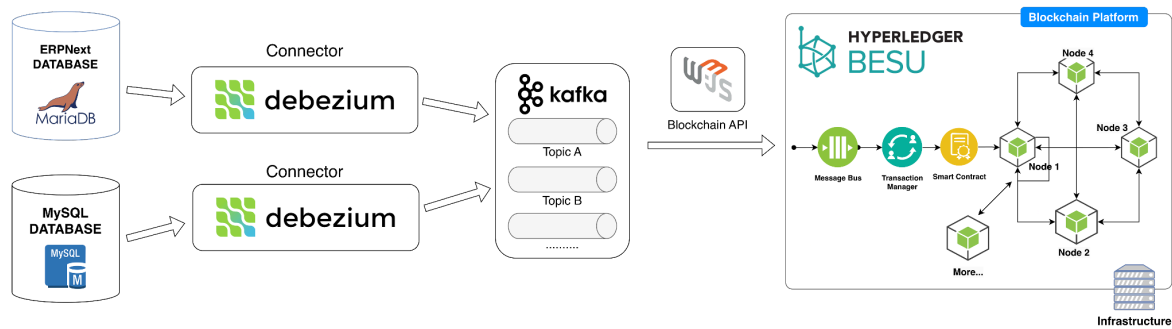


Figure 2. CDC Flow Diagram to Blockchain

2.5.1. Debezium Connector

Debezium is deployed as a Kafka Connect source connector to monitor the MariaDB binary log (binlog) in real-time. The connector is configured to track row-level changes (INSERT, UPDATE, DELETE) in the `tabEmployee` and `tabAttendance` tables within the ERPNext database. When a change occurs, Debezium captures the complete before-and-after state of the affected row, serializes the change event into JSON format, and publishes it to the corresponding Kafka topic. The connector configuration specifies `binlog.row.image=FULL` to ensure that all column values are included in the change event, enabling accurate hash computation for blockchain anchoring. Debezium's snapshot capability also allows for initial synchronization of existing records when the system is first deployed.

2.5.2. Apache Kafka

Apache Kafka serves as the distributed message broker that decouples the ERP transactional layer from the blockchain anchoring layer. Debezium publishes captured database events to specific Kafka topics following the naming convention:

```
erpnext.<database_name>.<table_name> (e.g., erpnext.frontend.tabAttendance).
```

A custom consumer service written in Node.js subscribes to these topics and processes the change events asynchronously. This architectural decoupling provides several critical benefits: if the blockchain network experiences temporary congestion or validator node failures, Kafka buffers the pending messages without blocking ERP operations; the consumer can implement retry logic with exponential backoff for failed blockchain transactions; and multiple consumer instances can be deployed for horizontal scaling during high-volume periods such as shift changes. The Kafka cluster is deployed using Confluent Platform containers with Zookeeper for cluster coordination, and a Kafka UI is provided for monitoring topic throughput and consumer lag.

2.6. System Architecture

The system is deployed on a development laptop running Ubuntu OS with Docker Desktop, utilizing hardware specifications of 16 CPU cores, 32 GB RAM, and 2 TB storage. All components are containerized using Docker to ensure environment isolation, reproducibility, and simplified deployment management. For more details see on the Figure 3.

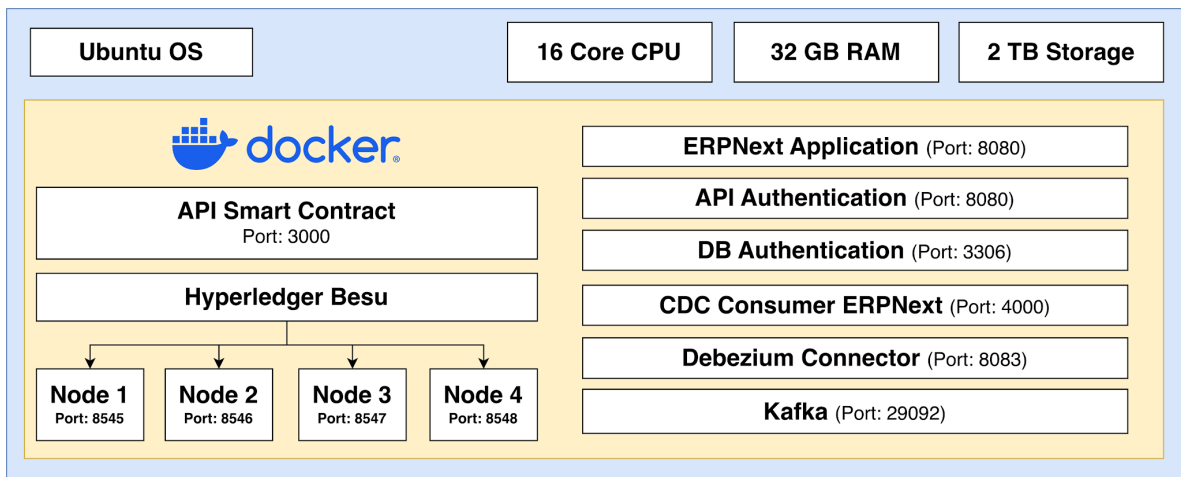


Figure 3. System Architecture

3. RESULTS AND DISCUSSION

This section evaluates the implementation of the integrated ERP-Blockchain platform and analyzes its performance across the data synchronization pipeline. Our evaluation focuses on three primary areas: the functionality of the custom-developed Frappe applications, the end-to-end latency of the event-driven Change Data Capture (CDC) mechanism, and the scalability of the Hyperledger Besu network under various consensus configurations.

3.1. ERPNext Custom Application Implementation

The custom application was successfully deployed on the Frappe framework. The modifications allowed for seamless UI integration where the blockchain status is visible to the HR manager.

3.1.1. Doctype

We made two custom Frappe applications were developed to extend ERPNext's HRM capabilities with IoT and blockchain integration. The system integrates an IoT module for real-time data acquisition tracking biometric events and time-series sensor data through dedicated device registries with a Blockchain module that anchors this information for permanent verification. By centralizing smart contract ABIs and recording transaction hashes, confirmation statuses, and gas metrics directly within the ERP, the platform provides a transparent audit trail that ensures data integrity for HR administrators without requiring external technical tools; for more details see in the table 3.

Table 3. IoT & Blockchain Module

Doctype	Purpose	Key Fields
IoT Device	Master data for IoT sensors/cameras	device_name, device_type (Sensor/Camera/Gateway), mac_address, ip_address, location, manufacturer, model, firmware_version, status, last_seen
IoT Sensor Data	Stores raw sensor readings	device (Link), sensor_type (Temperature/Humidity/Motion/Face), timestamp, value, unit, quality, raw_data (JSON)
Smart Contract	Registry of deployed contracts	contract_name, contract_address, chain_network, abi (JSON), status, description
Blockchain Transaction	Logs all blockchain writes	transaction_hash, from_address, to_address, timestamp, status (Pending/Confirmed/Failed), gas_used, smart_contract (Link)

3.1.2. Workspace

A dedicated "Blockchain Audit" workspace was created. This allows IT auditors to view the health of the connection between the ERP and the Blockchain network, ensuring the CDC services are running.

Table 4. Workspace Configuration

Module	Workspace Name	Sections
--------	----------------	----------

IoT	"Internet of Things"	IoT Dashboard (custom block), Shortcuts (IoT Device, IoT Sensor Data), Master Cards (Devices, Sensor Data, Settings)
Blockchain	"Blockchain"	Shortcuts (Smart Contract, Blockchain Transaction), Master Cards (Smart Contracts, Transactions)

Each module implements a dedicated workspace that organizes related functionality into a cohesive user interface. The IoT workspace features a real-time dashboard displaying device connectivity status and recent sensor readings, with quick-access shortcuts to frequently used DocTypes. The Blockchain workspace provides immediate access to contract management and transaction audit logs, structured with card-based navigation following Frappe's UX conventions. These workspaces integrate seamlessly with ERPNext's existing HR module, appearing in the sidebar navigation alongside standard HRM functions.

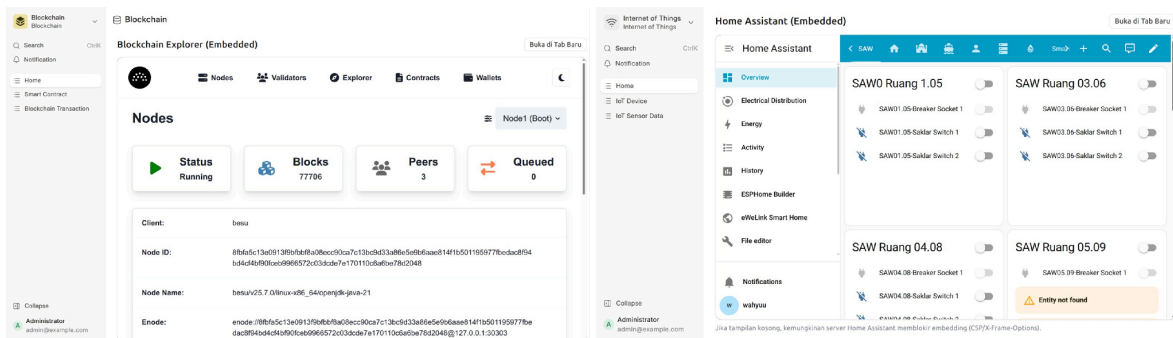


Figure 4. Custom Workspace of Blockchain & IoT

3.2. Real-time Data Record with Blockchain and Change Data Capture

We conducted performance testing to measure the latency of the entire pipeline. The test measured the time from the INSERT command in the ERP database to the final block confirmation on Hyperledger Besu. As shown in Table 5 and Figure 5., the CDC pipeline adds minimal overhead. The breakdown of latency for processing 1 to 250 records is as follows:

Table 5. CDC Pipeline Performance by Record Count

Data Entries (event)	DB to Kafka (s)	Kafka to Consumer (s)	Queue Time (s)	Consumer to Blockchain (s)	Total Time (s)
1	0.83	0.001	0.001	2.00	2.83
10	0.82	0.001	5.0	2.00	7.81
50	0.81	0.001	25.0	1.99	27.81
100	0.80	0.001	50.0	2.00	52.80
150	0.83	0.001	75.0	1.98	77.79
200	0.82	0.001	100.0	2.00	102.83
250	0.84	0.001	125.0	1.99	127.82

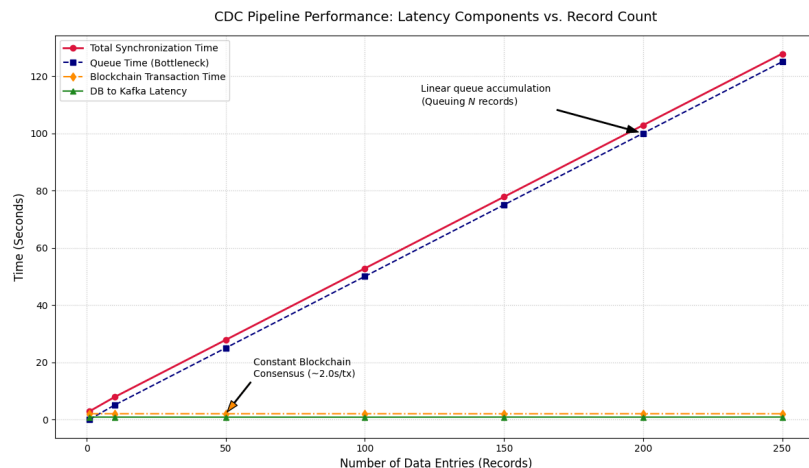


Figure 5. CDC Pipeline Performance Graph

The performance analysis reveals distinct behavioral patterns across pipeline stages. *DB to Kafka* latency remains consistently around 0.8 seconds regardless of record volume, demonstrating Debezium's efficient binary log parsing and event serialization. *Kafka to Consumer* delivery time is sub-millisecond (~1ms), confirming that Apache Kafka's message brokering introduces negligible overhead the streaming infrastructure is not a bottleneck.

The *Queue Time* metric warrants particular attention as it grows linearly with record count. This phenomenon occurs because blockchain write operations are inherently sequential each transaction must be mined into a block before the next can be confirmed. With a ~2 second blockchain confirmation time per record, processing N records creates a queuing delay of approximately $N \times 2$ seconds. This is not a limitation of Kafka (which can handle thousands of messages per second) but rather a consequence of blockchain's consensus mechanism ensuring transaction ordering and finality.

The *Total Time* formula accounting for queue accumulation is:

$$T_{total} = T_{db \rightarrow kafka} + T_{queue} + T_{kafka \rightarrow consumer} + T_{consumer \rightarrow blockchain}$$

For bulk operations, this can be approximated as:

$$T_{total}(N) \approx 0.82 + (N \times 2.0) + 0.001 + 2.0 \approx N \times 2.0 \text{ seconds}$$

This linear scaling is acceptable for batch synchronization scenarios but suggests optimization opportunities through transaction batching or parallel processing for high-volume deployments.

3.3. Evaluating Blockchain Performance via Validator Node Scaling

To ensure the scalability of the system, we analyzed how the configuration of the Hyperledger Besu network impacts performance. A critical aspect of private blockchain design is the number of validator nodes. While increasing nodes enhances security and decentralization, it increases the communication overhead required for consensus.

Table 6. IBFT Performance vs Validator Node Count

Nodes	Block Time (s)	TPS (tx/s)	Avg Latency (s)	Finality (s)	Success Rate (%)
2	N/A	N/A	N/A	N/A	Consensus fails
3	1.8	55	2.1	1.8	98.5
4	2.0	50	2.3	2.0	99.2
5	2.2	46	2.5	2.2	99.5
6	2.5	42	2.8	2.5	99.7
7	2.8	38	3.1	2.8	99.8
8	3.2	34	3.5	3.2	99.9
10	4.0	28	4.2	4.0	99.9

The experimental results demonstrate a clear inverse relationship between validator count and network throughput. IBFT consensus requires $2f+1$ validators for quorum, where f represents the maximum Byzantine faults tolerable. With 2 validators, the network cannot achieve consensus as it lacks the minimum 3-node quorum. At 3-4 nodes, the network operates optimally with 50-55 TPS and sub-3-second finality.

As validator count increases, message complexity grows quadratically ($O(n^2)$) because each validator must exchange prepare/commit messages with all peers. This introduces network latency and reduces block production frequency. However, higher node counts provide increased fault tolerance a 10-node network tolerates up to 3 Byzantine failures while maintaining 99.9% success rate.

4. CONCLUSION

This study successfully demonstrated the development of a customizable ERP system for Human Resource Management integrated with a private blockchain network using the Frappe Framework and an event-driven CDC architecture (Debezium and Apache Kafka). Experimental results reveal that while the streaming infrastructure introduces negligible overhead with sub-millisecond Kafka-to-Consumer delivery, the primary bottleneck resides in the Consumer-to-Blockchain stage (averaging 2.0s per record) due to the

consensus mechanism required for transaction finality. Total synchronization for individual records averages approximately 2.83 seconds, though bulk operations experience linear queue accumulation, with 250 records requiring roughly 128 seconds. Blockchain performance analysis confirms that an optimal configuration of 4–6 validator nodes achieves 42 - 50 TPS with sub-3-second finality while maintaining 99.2 - 99.7% success rates. This system effectively bridges the gap between flexible ERP operations with future work focused on implementing transaction batching and automated IoT data capture to further enhance operational efficiency.

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AUTHOR CONTRIBUTIONS STATEMENT

This journal uses the Contributor Roles Taxonomy (CRediT) to recognize individual author contributions, reduce authorship disputes, and facilitate collaboration.

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C : Conceptualization

M : Methodology

So : Software

Va : Validation

Fo : Formal analysis

I : Investigation

R : Resources

D : Data Curation

O : Writing - Original Draft

E : Writing - Review & Editing

Vi : Visualization

Su : Supervision

P : Project administration

Fu : Funding acquisition

CONFLICT OF INTEREST STATEMENT

Author state no conflict of interest

INFORMED CONSENT

We have obtained informed consent from all individuals included in this study.

DATA AVAILABILITY

The data that support the findings of this study are not publicly available and can only be obtained upon reasonable request from the corresponding author.

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